THE TANGENT SPACE AT A POINT IS THE DUAL OF THE FIBRE OF THE MODULE OF RELATIVE DIFFERENTIALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written; the only informal phrase is 'the reader sees that a to D(a) is a k-derivation', which states a verification that is immediate from the multiplicativity of phi and is reproduced as written.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-tangent-space-cotangent-space is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- The fibre of the relative differentials is unchanged by base change to the fibre. Morphisms, Lemma morphisms-lemma-base-change-differentials, invoked at proof lines 3038-3039. Supplies that the right hand side depends only on the scheme theoretic fibre.
- The spectrum of a local ring represents the local behaviour of the tangent space. Schemes, Lemma schemes-lemma-morphism-from-spec-local-ring, invoked at proof lines 3040-3042. Supplies that replacing X by the spectrum of its local ring does not change T_{X/S,x}.
- The stalk of the sheaf of differentials at a point is the module of differentials of the local ring. Modules, Lemma modules-lemma-stalk-module-differentials, invoked at proof lines 3043-3044. Supplies that the same replacement does not change the fibre of Omega_{X/S}.
- Universal property of the module of differentials. Standard algebra, used at proof lines 3053-3055. Supplies the bijection between k-derivations and kappa-linear maps on the module of differentials.

Source reference. Original lemma 08TE: sources/stacks-chow/tags.txt, source lines 11162–11162. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01V0: sources/stacks-chow/tags.txt, source lines 2429–2429. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01J6: sources/stacks-chow/tags.txt, source lines 2003–2003. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$. There is a canonical isomorphism*

$$T_{X/S,x} = \mathrm{Hom}_{\mathcal{O}_{X,x}}(\Omega_{X/S,x}, \kappa(x))$$

of vector spaces over $\kappa(x)$.

Proof. Set $\kappa = \kappa(x)$. Given $\theta \in T_{X/S,x}$ we obtain a map

$$\theta^*\Omega_{X/S} \to \Omega_{\mathrm{Spec}(\kappa[\epsilon])/\mathrm{Spec}(\kappa(s))} \to \Omega_{\mathrm{Spec}(\kappa[\epsilon])/\mathrm{Spec}(\kappa)}$$

Taking sections we obtain an $\mathcal{O}_{X,x}$-linear map $\xi_\theta : \Omega_{X/S,x} \to \kappa \mathrm{d}\epsilon$, i.e., an element of the right hand side of the formula of the lemma. To show that $\theta \mapsto \xi_\theta$ is an isomorphism we can replace S by s and X by the scheme theoretic fibre X_s. Indeed, both sides of the formula only depend on the scheme theoretic fibre; this is clear for $T_{X/S,x}$ and for the RHS see Morphisms, Lemma 01V0. We may also replace X by the spectrum of $\mathcal{O}_{X,x}$ as this does not change $T_{X/S,x}$ (Schemes, Lemma 01J6) nor $\Omega_{X/S,x}$ (Modules, Lemma 08TE).

Let $(A, \mathfrak{m}, \kappa)$ be a local ring over a field k. To finish the proof we have to show that any A-linear map $\xi : \Omega_{A/k} \to \kappa$ comes from a unique k-algebra map $\varphi : A \to \kappa[\epsilon]$ agreeing with the canonical map $c : A \to \kappa$ modulo ϵ. Write $\varphi(a) = c(a) + D(a)\epsilon$ the reader sees that $a \mapsto D(a)$ is a k-derivation. Using the universal property of $\Omega_{A/k}$ we see that each D corresponds to a unique ξ and vice versa. This finishes the proof. $\square$